IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 29 August 2023, accepted 6 September 2023, date of publication 11 September 2023, date of current version 14 September 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3313602

RESEARCH ARTICLE

Soft-Label for Multi-Domain Fake News Detection

DAOKANG WANG, WUBO ZHANG, WENHUAN WU, AND XIAOLEI GUO
Electrical and Information Engineering Department, Hubei University of Automotive Technology, Shiyan 442002, China
Corresponding author: Wubo Zhang (zhangwubo1234@126.com)

This work was supported in part by the Science and Technology Research Project of Department of Education of Hubei Province under Grant Q20201801, in part by the Natural Science Foundation of Hubei Province of China under Grant 2022CFB538, and in part by the Ph.D. Research Startup Fund Project of the Hubei University of Automotive Technology under Grant BK202004.

ABSTRACT The spread of fake news across several fields has had serious negative impacts on the public and society. Existing studies have shown that the use of multi-domain labels can improve the accuracy of fake news detection models since news from different domains has different characteristics. However, the previous multi-domain strategy has a problem: if the data lacks domain labels, the domain knowledge learned by the model for each domain cannot be used to determine whether the news is true or fake. Therefore, a novel multi-domain fake news detection model (SLFEND) is proposed to address the above problems by using soft labels. First, using our proposed Leap GRU to skip useless words, the membership function module generates soft labels for each news item. Then, multi-domain features of the messages are extracted using the soft labels to obtain the final overall feature representation. Finally, the overall feature set of the news is fed into the discriminator, and a judgment is made as to whether the news is real or fake. On the Weibo21 and Thu datasets, our proposed SLFEND approach achieves F1 scores of 92.49% and 89.98% respectively, surpassing existing state-of-the-art research. Experiments on the Weibo21 dataset demonstrate the SLFEND model's practicality and efficacy. Experiments using the Thu dataset indicate that SLFEND can transfer knowledge to the news without multi-domain labels and add suitable multi-domain labels to more scientifically determine whether the news is real or fake.

INDEX TERMS Soft label, fake news detection, multi-domain, leap GRU, expert panel.

I. INTRODUCTION

With the increasing popularity of the Internet, social media has grown rapidly in recent years due to its real-time convenience, openness, and two-way features, and has become one of the most important platforms for the public to receive news [1], [2]. At the same time, however, the low threshold of social media also encourages the growth and spread of false information, especially false news, in cyberspace. Especially during the COVID-19 pandemic [3], some self-media and netizens published a lot of unconfirmed news for the sake of traffic, which was then blindly followed and reprinted to expand the scope of dissemination, which not only hurt the audience but also damaged the authority and credibility of the mainstream media. But it has also created economic, political, and other hidden dangers [4]. Some news is difficult to distinguish accurately between true and false, even manually, and it is time-consuming and labor-intensive [5], [6]. This is why research into the detection of fake news in social media is so important today [7].

Social media platforms generate a large amount of news information every day and categorize it into news in different domains [8], [9] (e.g., social, political, and military). The challenge of recognizing fake news in many domains is referred to as multi-domain fake news detection (MFND) [10]. However, news from different domains suffers from a phenomenon called domain shift, where the use of words varies from domain to domain. For example, ''policeman'' and ''friend'' are commonly used in social news, while ''star'' and ''fan'' are commonly used in entertainment news, which reduces the versatility of the model [11], [12]. Furthermore, if there is insufficient news data on a given topic, the accuracy of the model will suffer. The multi-domain fake news detection model (MDFEND [13])

The associate editor coordinating the review of this manuscript and approving it for publication was Junhua Li.

adds appropriate single-domain labels to each news item as input, which can not only extract common features of different domains but also distinguish specific features of a single domain. A message can have the characteristics of multiple domains.FuDFEND [14] provides a fuzzy inference approach to offer a large number of labels for news items, which can more precisely describe the domain characteristics of each news, allowing the model to extract the characteristics of different fields of each news.

We build on this work by identifying two problems with FuDFEND. (1) When constructing multi-domain tags, the text of the input news is always read a large number of words are redundant, and skipping them does not affect performance. (2) When extracting news features, there is only one expert in each field, which has too much influence on the final result. To this end, we propose a soft-label fake news detection model (SLFEND) based on FuDFEND. Inspired by the Leap LSTM [15], we modify the standard GRU to dynamically jump between words when reading text. At each step, we use multiple feature encoders to extract news from the previous text, the following text, and the current word, and then decide whether to skip the current word to improve reading efficiency. Several experts in the same field form an expert group to extract news features to avoid subjective bias and blind spots in the judgment process. The main contributions of this paper can be summarised as follows:

- We propose a novel soft-label multi-domain fake news detection model (SLFEND), which extracts multi-domain features of news through generated soft labels.
- The standard GRU is modified and Leap GRU is proposed to skip unimportant word vectors when reading text vectors. Experimental results show that Leap GRU has a better predictive ability than GRU. To avoid subjective biases and blind spots in the judgment process of a single expert, the expert group system is used in the multi-domain fake news detection task.
- We ran a significant number of trials on two public Chinese datasets and the results showed that SLFEND outperformed other models in terms of indicators.

The remainder of the paper is organized as follows: Section II compiles a list of relevant work on false news identification from various angles. Section III introduces the proposed model and describes in detail the model structure and important components. The appropriate experiments are then carried out on the Weibo21 and Thu datasets in Section IV, and the experimental findings are examined and commented on. Finally, we summarize the findings, make inferences, and explore potential future study possibilities.

II. RELATED WORK

Fake news usually consists of some false, exaggerated, or inaccurate information that is intended to misinform the public or create misleading effects in order to achieve certain personal or political goals. In this part, we look at the various methods for spotting false news.

A. FAKE NEWS DETECTION

1) CONTENT-BASED METHODS

Content-based fake news detection is currently the most widely used method, mainly by analyzing the semantics, emotion, logic, and other characteristics of news content to judge the authenticity of the news. Based on labels and text attributes, Vyas et al. [16] use LSTM to determine whether messages are real or fake. The DFP method proposed by Raza et al. [17] mainly judges the credibility of information based on the content of the deepfake. Nasir et al. [18] proposed a hybrid CNN-RNN model that combines RNN and CNN to improve the detection performance. Saeid [19] proposed a WOX-Xgbtree algorithm for detection by analyzing and extracting content features. Long et al. [20] integrated the content of text and video and proposed an end-to-end multi-level multimodal cross-attention network for news detection.

2) USER INFORMATION-BASED METHODS

This method focuses on obtaining information from the behavior and characteristics of news dissemination and reading to judge whether the news is true or false. Dou et al. [21] proposed a new framework (UPFD) using user preferences and extracted information contained in user preferences from text and images simultaneously. Ahmadreza et al. [22] combined user comments and news and proposed a reinforcement adaptive learning fake news detection model, REAL-FND, which uses generalized and domain-independent features to judge whether the news is true or false. Zhu et al. [23] proposed the Entity Debiasing Framework (ENDEF) to mitigate entity bias from a causal perspective.

3) PROPAGATION-BASED METHODS

This method focuses on extracting information from features such as the propagation path and speed of news to determine whether the news is true or false. Sarith et al. [24] provide rich representations of social users involved in the spread of fake news, combined with propagation-based methods for fake news detection. Song et al. [25] fuse structure, content semantics, and temporal information and propose a temporal evolution graph neural network (TGNF) for fake news detection. Verma et al. [26] combine the propagation mode and user characteristics to propose a detection model ProFND and believed that the propagation time of real news is longer compared to fake news.The methods based on user information and dissemination have some limitations: data ethics issues, data loss and data noise issues [27]. Content-based approaches, on the other hand, are not as constrained and may identify and determine whether the news is real or fake in the absence of any additional information [28]. However, most studies are conducted in single or mixed domains. As a result, we are primarily interested in the application of content-based methodologies to the challenge of multi-domain false news identification.

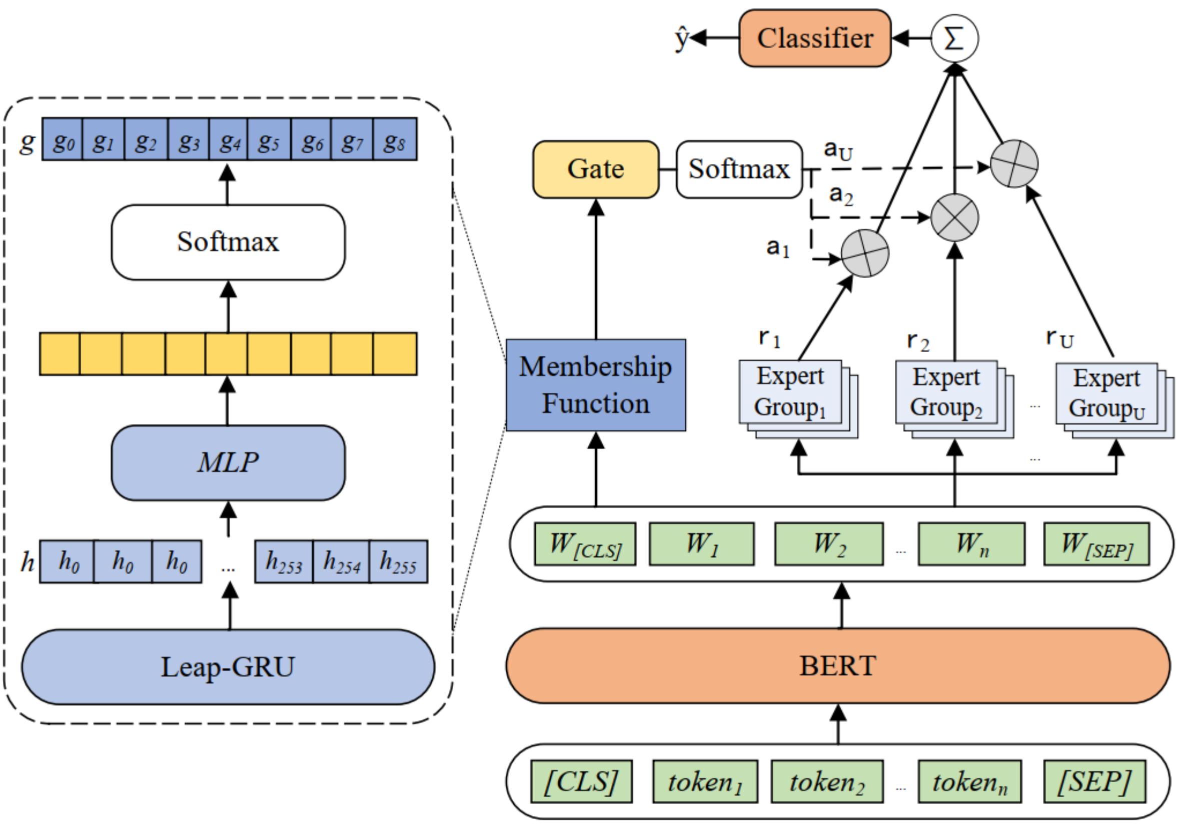


FIGURE 1. Overall framework of SLFEND.

B. MULTI DOMAIN LEARNING

Recent studies have shown that multi-domain learning is effective in detecting fake news. Yu et al. [29] created a framework for historical news environment perception, constructed historical and mainstream news environments for news, and fused them to improve multi-domain fake news detection. Nguyen et al. [30] proposed multi-domain and multimodal fake news detection model, V3MFND, uses multimodality to solve the problem of data sparsity. Zhu et al. [31] modeled the semantics, emotion, and style of news and proposed a memory-guided multi-view multi-domain fake news detection framework (M3FEND) to enrich in-domain information. The knowledge graph-based multi-domain fake news detection framework (KG-MFEND) proposed by Fu et al. [32] mitigates the domain differences at the word level by integrating external knowledge. It also adds label smoothing during training to reduce the impact of noise. The dual dynamic graph convolutional network (DDGCN) proposed by Sun et al. [33] models the dynamics in news dissemination and the background knowledge dynamics in the knowledge graph, and then combines with the time fusion unit to capture the cascade effect, thereby improving accuracy. In order to pursue a more reasonable and efficient multi-domain fake news detection technology, Nan et al. [13] had experts in several domains (TextCNN) score one news item and aggregate the findings through the 'domain gate' to get the final conclusion. The algorithm, however, is significantly reliant on human-annotated datasets, which must not only include true and false news but also know which category each news item belongs to. The domain label of a news item is not absolute and can have the characteristics of different domains. Limiting to a single label will reduce the performance of the model. On this basis, Liang et al. [14] proposed the FuDFEND model, which constructs multi-domain labels for each news by GRU, the feature extraction module obtains the final total feature vector by extracting the features of multi-domain labels, and then uses the discriminator to judge whether the news is true or false. This solves the above problems. However, the GRU needs to read the full text, which is inefficient when dealing with long news; there may be subjective biases and blind spots in the judgment process of a single expert in the same field. To solve this problem, we individually tweak the GRU and Expert modules and propose the SLFEND paradigm.

III. METHODOLOGY

In this chapter, we will go through the basic modules and operation of the proposed SLFEND model, the general framework of which is shown in Figure 1.

Feed news text into BERT [34] to obtain word embeddings $\boldsymbol{W} = [w_{[CLS]}, w_{[1]}, w_{[2]}, \ldots, w_{[n]}, w_{[SEP]}]$.The input $\boldsymbol{W}$ is processed and judged by experts in each area, and the opinions

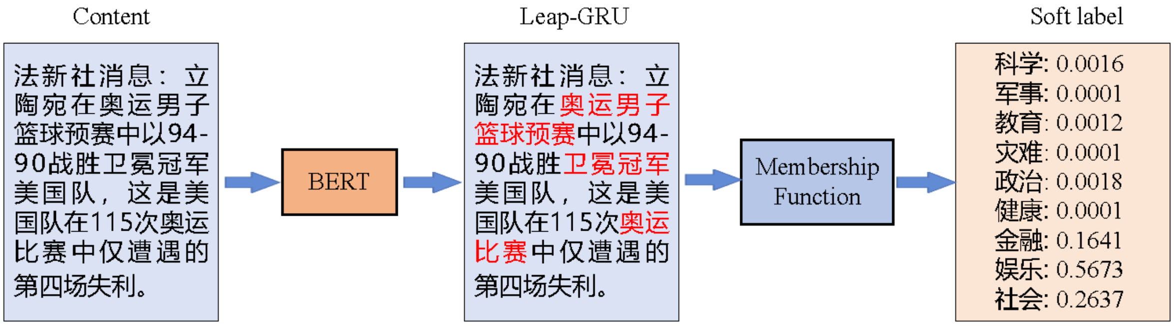


FIGURE 2. The operation process of the membership function.

of the experts in the group are summarized and fused as the judgment of the information in that domain. $\boldsymbol{W}$ is also sent via the membership function to obtain soft labels. The soft labels are subsequently sent via the domain gate and assigned the category weight $\boldsymbol{\alpha}$ to which they belong. It is then weighted and added to the findings of the associated expert group's judgment to produce the final feature vector $\boldsymbol{v}$ of the news. Finally, the final judgment result is obtained by feeding the feature vector $\boldsymbol{v}$ to the classifier.

A. MEMBERSHIP FUNCTION

The membership function consists of Leap-GRU, Multilayer Perceptron (MLP [35]), and Softmax function. We employ the membership function as a news multi-classification module throughout the training phase, assigning relevant category labels to each news. We only utilize it as a membership function in the usage phase to show the likelihood that each news item corresponds to each label. In particular, the news word embedding vector $\boldsymbol{W}$ is fed into Leap GRU, the useless word embedding vector is eliminated, and the appropriate output $\boldsymbol{h}$ is produced. The vector $\boldsymbol{g}$ is generated after normalizing $\boldsymbol{h}$ using the MLP and Softmax functions. The soft label we want is vector $\boldsymbol{g}$, which is made up of nine label vectors with dimensions that total up to one. Each label vector reflects the likelihood that the news(science, disaster, military, education, health, politics, finance, entertainment, and society) falls into each category. $\boldsymbol{M}(.;.)$ is the membership function representation, $\boldsymbol{\theta}$,$\boldsymbol{\theta_1}$,$\boldsymbol{\theta_2}$ are membership functions, Leap-GRU, parameters in MLP. $\boldsymbol{g}$ means soft label:

$$g = m(W; \theta) = softmax(MLP(LeapGRU(W; \theta_1); \theta_2)) \quad (1)$$

Figure 2 shows the execution process of the membership function, the example is randomly selected from Sina Weibo. The content of this message is "AFP reports that Lithuania defeated the US 94-90 in an Olympic men's basketball qualification game. Basketball preliminary round game, the reigning champions' fourth in 115 Olympic Games". It can be seen that Leap GRU only retained "Olympic men's basketball", "game", "defending champions" and other important words, leaving out the most unimportant words, and after analyzing the remaining steps of the membership function, it is finally concluded that the news has entertainment domain, society domain and characteristics of a financial domain. The details of Leap-GRU's work are described in the next section.

B. LEAP GRU

GRU always reads all input content, and the long sequence input process is slow. Removing most of the extraneous terms in a news story has little effect on the model's results in the evolving task of identifying false news [36]. Leap-GRU skips irrelevant information during text input and has better prediction results in terms of efficiency. Figure 3 depicts a Leap GRU process running to t. In the illustration, there are two circles: one grey for skipping this step and one orange for running GRU regularly. This example skips step h_t and proceeds to step x_{t+1} normally. Therefore, it copies h_{t-1} as ht and uses the standard GRU update function GRU(h_t, x_{t+1}) to get h_{t+1}.

From the word embedding $\boldsymbol{W}$, the Leap-GRU input sequence $[x_1,x_2 \ldots ,x_T]$ of length T and dimension d is obtained or called $x_{1:T}$, where $x_t \in \mathbb{R}^f$ is the t-th element of the word vector. In item t, the features of the forward sequence $(x_{1:t-1})$ are encoded as $f_{precede}(t) \in \mathbb{R}^p$ and the features of the backward sequence $(x_{t+1:T})$ are encoded as $f_{follow}(t) \in \mathbb{R}^f$. This paper divides the backward sequence into two parts: local backward text and global backward text. For example, the t-th item x_t,$x_{t+1:t+m}$ in x1:T is a local backward sequence encoded by CNN. $x_{t+1:T}$ is the global backward sequence. To extract the latter sequence, this paper uses a miniature reverse GRU, which reads the last element of the sequence in reverse order. Let GRU(t') and CNN$(t')$of the reverse GRU and CNN be the output of step t, respectively. Among them, GRU(t') encodes the features of $x_{t':T}$, and CNN(t) encodes the features of $x_{t':t+m}$. The features of the reverse sequence are as follows.

$$f_{follow}(t) = \begin{cases} GRU(t+1); CNN(t+1), & t < T \\ h_{end}, & t = T \end{cases} \quad (2)$$

where $h_{end} \in \mathbb{R}^f$ indicates that the sequence must stop when it reaches the end, and h_{end} and other model parameters must be learned simultaneously during training. We decide to utilize the likelihood given by MLP to determine whether

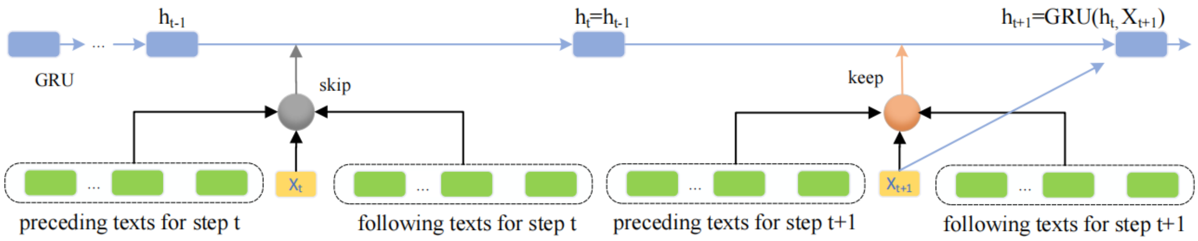


FIGURE 3. **The operation process of the membership function.**

a word vector should be skipped or processed normally. MLP employs two layers.

$$s_t = RELU(\hat{W}_1[x_t; f_{preced}(t); f_{follow}(t)] + b_1) \quad (3)$$

$$\pi_t = Softmax(\hat{W}_2 s_t + b_2) \quad (4)$$

Among them, $\hat{W}_1 \in \mathbb{R}^{s*(d+p+f)}$, $\hat{W}_2 \in \mathbb{R}^{2*s}$, $b_1 \in \mathbb{R}^s$, $b_2 \in \mathbb{R}^2$ are the weights and bias values of the multi-layer perceptron. ";" is the connection symbol, $s_t \in \mathbb{R}^s$ represents the hidden layer's state, while $\pi_t \in \mathbb{R}^2$ indicates the probability value. For the probability value π_t, the threshold value chosen in this paper is 0.5. If $\pi_t \geq 0.5$, use the normal GRU to process a word vector; if $\pi_t < 0.5$, do not update the corresponding hidden layer and skip immediately without using this word vector.

$$h_t = \begin{cases} GRU(h_{t-1}, x_t) & \pi_t \geq 0.5 \\ h_{t-1} & \pi_t < 0.5 \end{cases} \quad (5)$$

Send the output h to the MLP to obtain a nine-dimensional vector that is generated then apply the Softmax method to normalize the output to obtain vector $\boldsymbol{g}$.

C. FEATURE EXTRACTION

We use mixed expert groups [37] to extract features from news content, where each expert group has three experts. In our model, each expert group consists of a TextCNN [38], a DCNN [39], and a DPCNN [40]. Experts can better extract the features of news in their industry and make more competent and authoritative judgments about it. The expert network can be expressed as $E_{ij}(.;.)(1 \leq i \leq U)$ $(1 \leq j \leq C)$. U and C are two hyperparameters representing the expert group and the number of experts in each expert group, respectively. $\boldsymbol{r_i}$ represents the output of the expert group. $\boldsymbol{\varphi_{ij}}$ is the trainable parameter of the expert:

$$r_i = \frac{1}{C}\sum E_{ij}(W; \varphi_{ij}) \quad (6)$$

D. DOMAIN GATE

Since different expert groups can extract features from different domains, we feed soft labels into domain gates to obtain weight scores. The expert group will be reasonably accurate in assessing news in its field, and other expert groups can add complementing references to the judgment findings. As a result, we assign weights $\boldsymbol{\alpha}$ to the expert groups' assessments. In this case, U is set to the same number 9 as the dimension of the vector $\boldsymbol{g}$, and $\boldsymbol{\psi}$ is some trainable parameter in the domain gate.

$$\begin{aligned} \alpha &= (\alpha_1, \alpha_2, \ldots, \alpha_U) = Gate(G; \psi) \\ &= Softmax(MLP(G; \psi)) \end{aligned} \quad (7)$$

E. FUNCTION OF LOSS

To produce the final vector $\boldsymbol{v}$, we weighted and summed the processing outcomes of all expert groups using varied weights $\boldsymbol{\alpha}$.

$$v = \sum_{i=1}^{U} \alpha_{ij} r_{ij} \quad (8)$$

The MLP uses the feature vector V to determine if the message is true or false. The MLP produces a probability value between 0 and 1, known as $\hat{y}$. The higher the $\hat{y}$ number, the more likely the message is judged false. The MLP is viewed as a binary classification approach with a sigmoid output layer in this context.

$$\hat{y} = sigmod(MLP(v, \xi)) \quad (9)$$

The letter $\boldsymbol{\xi}$ denotes trainable variables in MLP. We denote the true label value by y_i and the predicted value by $\hat{y}_i$. During training, use the binary cross-entropy loss function as the loss function:

$$L = \sum_{i=1}^{N}(y_i log\hat{y}_i + (1 - y_i) log(1 - \hat{y}_i)) \quad (10)$$

IV. EXPERIMENT

We conduct comprehensive tests on multi-domain and domain-free labeled datasets to evaluate the performance of our proposed SLFEND model. The experimental results provide answers to the three following questions:

- EQ1: Does our proposed SLFEND model outperform other baseline models?
- EQ2: What is the impact of each module of the SLFEND model on fake news detection results?
- EQ3: What is the transfer learning capability of the SLFEND model? How sensitive are the hyperparameters?

TABLE 1. Data Statistics of Weibo21.

Domain	Science	Military	Education	Disasters	Politics	Health	Finance	Entertainment	Society	all
Real	143	121	243	185	306	485	959	1000	1198	4640
Fake	93	222	248	591	546	515	362	440	1471	4488
Total	236	243	491	776	852	1000	1321	1440	2669	9128

A. DATASET

We evaluated the performance of our proposed SLFEND model on two Chinese test sets, weibo21 [13] and Thu [14].

The Chinese Academy of Sciences issued the weibo21 dataset, which is a multi-domain dataset. The collection contains 9128 news items, 4488 fraudulent news, and 4640 actual news. science, disaster, military, education, health, politics, finance, entertainment, and society is the nine categories of news. Manually label each news item with a category label, and the distribution of the number of news items in each category is displayed in Table 1:

The news in the Thu dataset has no domain labels and is a mixed-domain dataset. It consists of 1849 real news items and 1560 fake news items. Zero-shot learning is performed to test the model's ability to transfer domain knowledge.

B. BASELINE MODELS

To verify model performance, three types of baseline models are used for analysis. The first is a single domain baseline including TextCNN, BiGRU, and BERT. Each model is tested on news from only one domain at a time. The second is a mixed domain baseline, including BiGRU, TextCNN, and BERT. Each model is tested on news from all domains at the same time. Finally, multi-domain baselines are available, such as EANN models, MMoE models, MoSE models, EDDFN models, MDFEND models, and FuD-FEND models. Each model is described below:

TextCNN: We use 5 convolutional kernels with 64 channels and get the embedded input via word2vec.

BiGRU: BiGRU is a regularly used baseline for detecting false news. The news is recorded in chronological order to retain the sequence of information in the news.

BERT: Abandon the first 11 layers of Transformer blocks in BERT and use only the last layer of Transformer blocks to convert news into word embedding vectors and use MLP to judge whether it is real or fake.

EANN [41]: This is a multi-mode fake news detection method that uses only the text extraction part to extract domain-independent features. The model architecture is the same as that of TextCNN.

MMOE [42] and MOSE [43]: MMOE and MOSE are technique models in the field of multi-task learning research. We approach news identification in each domain as a separate job, handle the multi-domain problem as a multi-task problem, and identify false news using MMOE and MOSE.

EDDFN [44]: is a multi-domain fake news detection model that can model different domains and retain domain-specific knowledge. In our experiments, manually labels knowledge is used for fake news detection.

TABLE 2. The details of the parameters.

Parameter	Value	Parameter	Value
Batch size	16	Max epochs	10
Dropout	0.2	Embedding size	768
optimizer	Adam	Learning rate	2e-05
U	9	C	3

MDFEND: Experts in different domains extract news features, domain gates assign different weights to experts, and the total feature vector is obtained after aggregation.

FuDFEND: First, utilize the BERT model's final layer Transformer block to turn the news into word embedding vectors. which constructs multi-domain tags for each news by GRU, and the feature extraction module obtains the final total feature vector by extracting multi-domain tag features. By enhancing the MDFEND model, the model recognizes news without domain labels and utilizes the fuzzy reasoning method to designate multi-domain labels for news.

C. EXPERIMENT SETTING

We randomly split all data sets into training, validation, and test sets in a 4:1:1 ratio. All baselines follow the parameter settings in the original paper. To avoid differences in the effect of each model due to the experimental environment, all experiments are run on the same hardware and software: Ubuntu 18.04 system is used, memory is 16G, the processor is Core i7-12700H, the graphics card is NVIDIA Tesla P100 16GB, the development environment is python 3.9.16, deep learning framework is PyTorch 1.8.0. The MLP used in these models consists of a dense layer (384 hidden units). The maximum sentence length is set to 170, the dimension of the BERT word embedding vector is fixed at 768, and the dimension of the Word2Vec word embedding vector is 200. The learning rate is 2e-5. Table 2 shows the numerical values for several of the model's main parameters.

Evaluating Indicator To evaluate model effectiveness, accuracy (Acc), precision (P), recall (R), F1, and area under the ROC curve (AUC) are used as evaluation indicators.

Accuracy, which is the proportion of all predicted correct samples out of all predicted results:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

Precision rate, which is the proportion of positive samples correctly predicted as positive out of all predicted positive samples:

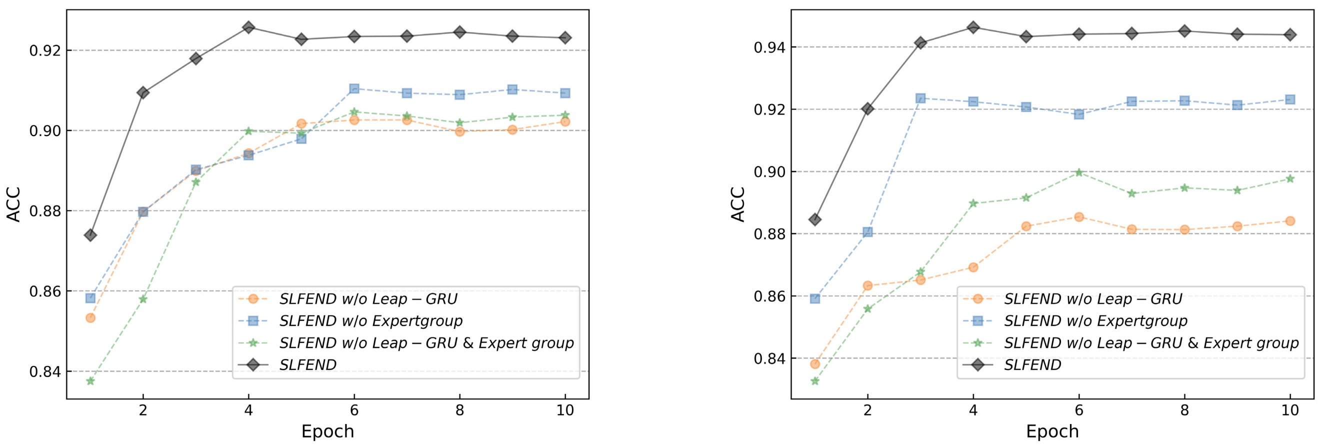
$$P = \frac{TP}{TP + FP} \tag{12}$$

TABLE 3. Experiments on Weibo21 dataset (F1-score).

	model	Science	Military	Education	Disasters	Politics	Health	Finance	Entertainment	Society	all
single	TextCNN	0.7470	0.778	0.8882	0.8310	0.8694	0.9053	0.7909	0.8591	0.8727	0.8380
	BiGRU	0.4896	0.7169	0.7067	0.7625	0.8477	0.8378	0.8190	0.8380	0.6067	0.7342
	BERT	0.8192	0.7795	0.8136	0.7885	0.8188	0.8909	0.8464	0.8638	0.8242	0.8272
mixed	TextCNN	0.7254	0.8839	0.8362	0.8222	0.8561	0.8768	0.8638	0.8456	0.8540	0.8686
	BiGRU	0.7269	0.8724	0.8138	0.7935	0.8356	0.8868	0.8291	0.8629	0.8485	0.8595
	BERT	0.7777	0.9072	0.8331	0.8512	0.8366	0.9090	0.8735	0.8769	0.8577	0.8795
multi	EANN	0.8225	0.9274	0.8624	0.8666	0.8705	0.9150	0.8710	0.8957	0.8877	0.8975
	MMOE	0.8755	0.9112	0.8706	0.8770	0.8620	0.9364	0.8567	0.8886	0.8750	0.8947
	MOSE	0.8502	0.8858	0.8815	0.8672	0.8808	0.9179	0.8672	0.8913	0.8729	0.8939
	EDDFN	0.8186	0.9137	0.8676	0.8786	0.8478	0.9379	0.8636	0.8832	0.8689	0.8919
	MDFEND	0.8301	0.9389	0.8917	0.9003	0.8865	0.9400	0.8591	0.9066	0.8980	0.9137
	FuDFEND	0.8133	0.9468	0.8917	0.9059	0.9013	0.9417	0.8901	0.9161	0.9174	0.9213
	SLFEND	0.8833	0.9234	0.9286	0.891	0.9370	0.9243	0.9233	0.9254	0.9265	0.9249
	Improvement	0.78%	-	3.69%	-	2.97%	-	3.32%	0.93%	0.91%	0.36%

TABLE 4. Experimental results on Weibo21 dataset. (F1-score).

model	science	military	Education	Disasters	Politics	Health	Finance	Entertainment	Society	all
w/o Leap-GRU	0.8712	0.9365	0.8990	0.8886	0.8792	0.9347	0.9063	0.9062	0.8993	0.9176
w/o Expert group	0.8407	0.9186	0.9182	0.8614	0.9227	0.9049	0.9307	0.9067	0.9141	0.9203
w/o Leap-GRU & Expert group	0.8418	0.9437	0.8990	0.8895	0.8789	0.9430	0.9106	0.8938	0.9013	0.9147
SLFEND	0.8833	0.9234	0.9286	0.891	0.9370	0.9243	0.9233	0.9254	0.9265	0.9249



(a) Ablation studies of society

(b) Ablation studies of politics

FIGURE 4. Single domain ablation experiments.

Recall rate, which is the proportion of positive samples correctly predicted as positive out of all positive samples:

$$R = \frac{TP}{TP + FN} \tag{13}$$

F1-score, F1-score, when P and R are considered simultaneously, both attain the greatest value at the same moment. this task improves the recall rate on the premise of ensuring the accuracy rate:

$$F_1 = \frac{2 \times P \times R}{P + R} \tag{14}$$

AUC measures the ability of a binary classification model to discriminate between positive and negative samples with high and low confidence. The AUC value ranges between 0.5 and 1, with the closer to 1, indicating more model fidelity.

$$AUC = \frac{1 + \frac{TP}{TP+FN} - \frac{FP}{TN+FP}}{2} \tag{15}$$

D. Weibo21 EXPERIMENT RESULTS

To answer the first question, we first run tests on the Weibo21 dataset and compare our model to the previously established baseline models. The exact experimental findings of the single-domain, mixed-domain, and multi-domain benchmark models are shown in Table 3.

As shown in Table 3, the SLFEND model's outcomes improved by 0.78%, 3.69%, 2.97%, 3.32%, 0.93%, and 0.91% in the six disciplines of science, education, politics, finance, entertainment, and society, respectively. when compared to the benchmark model's best categories, This

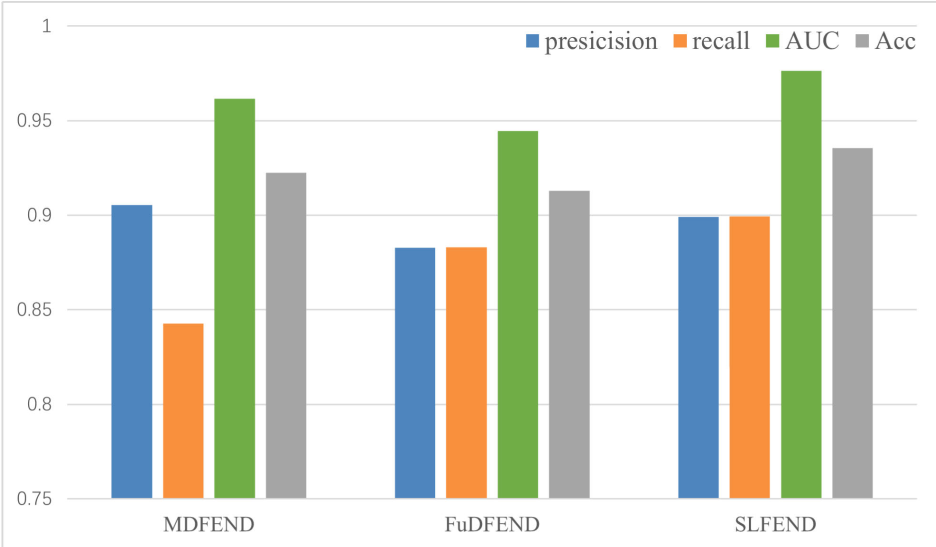


FIGURE 5. Experimental results of the Thu dataset.

demonstrates that our model is typically superior to the baseline model and provides an answer to the first question.

We find that mixed-domain methods outperform single-domain methods overall, suggesting that multi-domain joint training methods can improve domain-specific performance. We assume that this finding is because each piece of news not only includes the distinctive information of one domain, but single-domain news does not utilize the information of other domains, resulting in weaker outcomes than mixed domains. The comparison of outcomes is more visible in the case of imbalanced data and few single-domain data sets.

Among multi-domain methods, our SLFEND achieves better results. EANN, MMOE, MOSE, and EDDFN all employ sharing methods to acquire cross-domain shared knowledge, however, learning shared information with a common structure is challenging. MDFEND and FuDFEND employ a soft-sharing approach called knowledge embedding to combine useful entity knowledge. The fuzzy reasoning mechanism and hybrid expert method proposed by FuDFEND can effectively learn some unlabelled features. However, it is inefficient when dealing with longer messages, and there may be subjective biases and blind spots in the judgment process of a single expert in the same field. The model we propose can skip useless words to improve efficiency and form a group of experts to reduce personal subjective bias and blind spots.

E. ABLATION STUDY

An ablation experiment was carried out to evaluate the impact of each module of the SLFEND model on the detection outcomes of false news. Based on the basis of the main body, we made the following changes in this experiment:

"w/o Leap GRU" does not use the Leap GRU, but uses the standard GRU.

"w/o expert group" does not use the expert group method and uses only one TextCNN expert per domain.

TABLE 5. Society domain results of Weibo 21 dataset.

model	precision	recall	acc	auc	F1
w/o Leap-GRU	0.8994	0.9027	0.9026	0.9643	0.8993
w/o Expert group	0.9143	0.9025	0.9104	0.9647	0.9141
w/o Leap-GRU & Expert group	0.9011	0.9055	0.9045	0.9572	0.9013
SLFEND	0.9265	0.9213	0.9257	0.9631	0.9265

TABLE 6. Politics domain results of Weibo 21 dataset.

model	precision	recall	acc	auc	F1
w/o Leap-GRU	0.8741	0.8863	0.8854	0.9625	0.8792
w/o Expert group	0.9223	0.9139	0.9235	0.9621	0.9227
w/o Leap-GRU & Expert group	0.8715	0.8634	0.8996	0.9476	0.8789
SLFEND	0.9383	0.9635	0.9413	0.9749	0.9370

"w/o Leap-GRU & Expert Group" does not use both Leap-GRU and Expert Group methods.

Table 4 displays the outcomes of the model's ablation trials as a whole. We conducted ablation experiments in both the social and political arenas to better understand the function of specific modules. The results of the ablation experiments in the social domain are shown in Table 5, while the results of the ablation experiments in the political domain are shown in Table Table 6. The variations in the Acc value of each module in the two domains are depicted in Figure 4. We have the following comments:

The results show that the results of "w/o Leap-GRU & Expert group" is not as good as that of "w/o Leap-GRU", indicating that the judgment of a single domain expert group is more objective than that of a single expert. The performance of "w/o Leap-GRU & Expert group" is not as good as that of "w/o Expert group", indicating that Leap-GRU can effectively improve model performance. SLFEND achieved

TABLE 7. Experimental results on Thu dataset. (F1-score).

model	science	military	Education	Accidents	Politics	Health	Finance	Entertainment	Society	all
MDFEND	0.7949	0.7710	0.8565	0.6878	0.7902	0.8616	0.7885	0.8249	0.8503	0.8697
FUDFEND	0.8249	0.8697	0.8581	0.6971	0.8135	0.8393	0.7972	0.8339	0.8551	0.8825
SLFEND	0.8371	0.8635	0.9117	0.7475	0.8147	0.8850	0.8036	0.8634	0.8641	0.8998

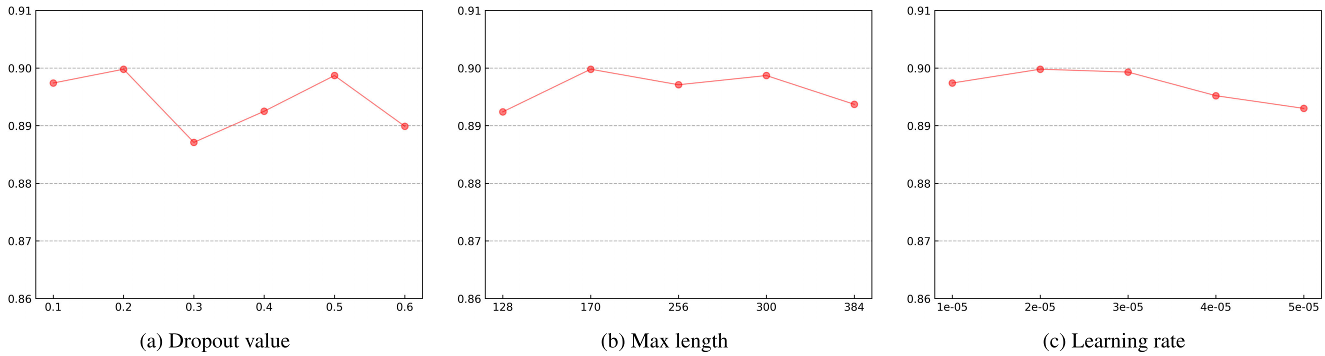


FIGURE 6. Performance (F1) of SLFEND with various hyperparameters.

the best results in the above experiments, indicating that Leap-GRU and Expert group are indispensable, and it also proves that the model is effective in multi-domain fake news detection.

According to the trend in Figure 4, it can be seen that SLFEND tends to be stable in the fourth round, while w/o Leap-GRU & Expert group converges only in the sixth epoch, which shows that the convergence speed of SLFEND is getting faster and faster. The w/o Expert group converges earlier than the other two cases. It can be concluded that the robustness of SLFEND can be improved by using Leap-GRU.

F. EXPERIMENT ON THU DATASET

To verify the transfer learning capability of SLFEND, we conducted experiments on the Thu dataset, selecting MDFEND and FuDFEND as benchmark models for reproduction. The messages in the original Thu dataset have no domain labels. To test MDFEND, a single-domain labeled version of FuDFEND is used, and MDFEND is trained using the hyperparameters given by nan [13]. The results are shown in Table 7 and Figure 5.

Table 7 shows that compared to the best single domain results in MDFEND and FuDFEND, science, education, politics, finance, entertainment, and society increased by 1.22%, 5.36%, 0.12%, 0.64%, 2.95% and 0.9% respectively. The results show that SLFEND is an effective method for multi-domain fake news detection. In Figure 5, we visualize the experimental results of Precision, Recall, AUC, and Acc. The precision value is slightly lower than MDFEND and higher than FuDFEND, indicating that the false positive rate of SLFEND is low. The recall value of SLFEND is significantly higher than the other two models, indicating that it has a wide coverage of fake news and a low false negative rate. The AUC value is the highest, indicating that the performance of SLFEND is relatively stable under different thresholds. The Acc value is the highest, indicating that the overall performance of SLFEND is the best. Overall, SLFEND performed better than all of the comparison models, indicating that our model has good transferability and a high reference value.

TABLE 8. The running time of an epoch.

Model	MDFEND	FUDFEND	LGFEND
Time(s)	52	73	57

TABLE 9. The running time of an epoch at different Max lengths.

Max length	128	170	256	300	384
Time(s)	48	57	91	149	334

G. HYPERPARAMETER EXPERIMENT

This section focuses on the influence of three hyperparameters - dropout value, Max length, and learning rate - on the outcomes of an experiment performed on a Thu dataset. Figure 6 depicts the F1-score when the three hyperparameters are changed. The maximum length of the comparison model was likewise specified to be the same as that of the SLFEND model for the sake of scientific fairness in the comparison test. Table 8 compares the average time required to complete an epoch for various models. Table 9 displays the average run time of an epoch for various SLFEND model lengths.

We observe that the SLFEND model performs best with a dropout value of 0.2, a maximum sequence length of 170, and a learning rate of 2e-05. In particular, on the Thu dataset, even with the worst setting (F1 of 0.8899), our model outperforms the best baseline (F1 of 0.8825), proving that the SLFEND model is not sensitive to the above hyperparameters [45].

TABLE 10. SOTA performance comparison table(F1-score).

	model	science	military	Education	Accidents	Politics	Health	Finance	Entertainment	Society	all
weibo21	FuDFEND [14]	0.8133	0.9468	0.8917	0.9059	0.9013	0.9417	0.8901	0.9161	0.9174	0.9213
	M3FEND [31]	0.8292	**0.9506**	0.8998	0.8896	0.8825	0.9460	0.9009	0.9315	0.9089	0.9216
	DITFEND [46]	0.8426	0.9419	0.9091	**0.9145**	0.8895	**0.9550**	0.8984	0.9100	0.8991	-
	SLFEND	**0.8833**	0.9234	**0.9286**	0.891	**0.9370**	0.9243	**0.9233**	**0.9254**	**0.9265**	**0.9249**
Thu	MDFEND [14]	0.8157	0.8407	0.7995	0.6878	0.7938	0.8364	0.7535	0.8047	0.8642	0.8731
	FuDFEND [14]	**0.8760**	0.8301	0.8568	0.7129	**0.8148**	0.8753	0.7930	0.8492	**0.8653**	0.8900
	SLFEND	0.8371	**0.8635**	**0.9117**	**0.7475**	0.8147	**0.8850**	**0.8036**	**0.8634**	0.8641	**0.8998**

From Figure 6b and Table 9, it can be seen that increasing the input length of the model does not necessarily improve the performance and may increase the model row time. As shown in Table 8, the running time of SLFEND is not the shortest of all the baselines. When the max length is 128, it takes only 48 seconds to run an epoch, which is the shortest time of all the lengths. Although the running time is shorter than when the max length is 170, the F1 score is relatively lower. Combining the above results, we concluded that the best results for SLFEND models were obtained when the super-parameter dropout value was 0.2, the max length was 170 and the learning rate was 2e-05.

H. STATE-OF-THE-ART COMPARATIVE EXPERIMENT

Table 10 compares the performance of our proposed SLFEND with other state-of-the-art studies on the weibo21 and Thu datasets. SLFEND generates superior F1 scores in various areas on both datasets, demonstrating the great performance of our model in dealing with fake news in these specific areas. The overall F1 scores reached the maximum values of 92.49% and 89.98%, respectively. The results suggest that our proposed novel strategy outperforms existing state-of-the-art research.

The three questions posed at the beginning of this chapter are answered through a series of experiments, demonstrating the progress and practicality of the SLFEND model. It not only offers a novel solution to the challenge of false news detection but also provides pointers and ideas for other researchers. It has broad application prospects.

V. CONCLUSION

In this paper, we propose a new fake news detection model, SLFEND. In SLFEND, we first use leap GRU to skip words that are unrelated to the prediction and improve operational efficiency. Then, multi-domain features of news are extracted using soft labels to obtain the final overall feature representation. Finally, the classification module with an MLP structure is employed to determine the veracity of the news. Experiments on the weibo21 data set illustrate the model's higher performance. Experiments using the Thu data set demonstrate the model's knowledge transfer capabilities and practicability.

In future studies, we will investigate false news identification from a broader perspective, incorporating news comments and the emotional propensity of news to increase detection accuracy.

REFERENCES

[1] N. Seddari, A. Derhab, M. Belaoued, W. Halboob, J. Al-Muhtadi, and A. Bouras, "A hybrid linguistic and knowledge-based analysis approach for fake news detection on social media," *IEEE Access*, vol. 10, pp. 62097–62109, 2022.

[2] M. Kang, J. Seo, C. Park, and H. Lim, "Utilization strategy of user engagements in Korean fake news detection," *IEEE Access*, vol. 10, pp. 79516–79525, 2022.

[3] M. Narra, M. Umer, S. Sadiq, A. A. Eshmawi, H. Karamti, A. Mohamed, and I. Ashraf, "Selective feature sets based fake news detection for COVID-19 to manage infodemic," *IEEE Access*, vol. 10, pp. 98724–98736, 2022.

[4] T. Jiang, J. P. Li, A. U. Haq, A. Saboor, and A. Ali, "A novel stacking approach for accurate detection of fake news," *IEEE Access*, vol. 9, pp. 22626–22639, 2021.

[5] P. Li, X. Sun, H. Yu, Y. Tian, F. Yao, and G. Xu, "Entity-oriented multi-modal alignment and fusion network for fake news detection," *IEEE Trans. Multimedia*, vol. 24, pp. 3455–3468, 2022.

[6] M. Amjad, G. Sidorov, A. Zhila, A. Gelbukh, and P. Rosso, "Overview of the shared task on fake news detection in Urdu at FIRE 2020," in *Proc. CEUR Workshop*, vol. 2826. CEUR-WS, 2020, pp. 434–446.

[7] P. Vyas, J. Liu, and O. El-Gayar, "Fake news detection on the web: An LSTM-based approach," in *Proc. AMCIS*, vol. 5, 2021.

[8] M. Davoudi, M. R. Moosavi, and M. H. Sadreddini, "DSS: A hybrid deep model for fake news detection using propagation tree and stance network," *Exp. Syst. Appl.*, vol. 198, Jul. 2022, Art. no. 116635.

[9] S. R. Sahoo and B. B. Gupta, "Multiple features based approach for automatic fake news detection on social networks using deep learning," *Appl. Soft Comput.*, vol. 100, Mar. 2021, Art. no. 106983.

[10] C. Song, N. Ning, Y. Zhang, and B. Wu, "Knowledge augmented transformer for adversarial multidomain multiclassification multimodal fake news detection," *Neurocomputing*, vol. 462, pp. 88–100, Oct. 2021.

[11] G. Shan, B. Zhao, J. R. Clavin, H. Zhang, and S. Duan, "Poligraph: Intrusion-tolerant and distributed fake news detection system," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 28–41, 2022.

[12] H. Chi and B. Liao, "A quantitative argumentation-based automated explainable decision system for fake news detection on social media," *Knowl.-Based Syst.*, vol. 242, Apr. 2022, Art. no. 108378.

[13] Q. Nan, J. Cao, Y. Zhu, Y. Wang, and J. Li, "MDFEND: Multi-domain fake news detection," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manag.*, 2021, pp. 3343–3347.

[14] C. Liang, Y. Zhang, X. Li, J. Zhang, and Y. Yu, "FuDFEND: Fuzzy-domain for multi-domain fake news detection," in *Natural Language Processing and Chinese Computing*. Guilin, China: Springer, 2022, pp. 45–57.

[15] T. Huang, G. Shen, and Z.-H. Deng, "Leap-LSTM: Enhancing long short-term memory for text categorization," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 1–7.

[16] P. Vyas and O. F. El-Gayar, "Credibility analysis of news on Twitter using LSTM: An exploratory study," in *Proc. AMCIS*, vol. 17, 2020.

[17] A. Raza, K. Munir, and M. Almutairi, "A novel deep learning approach for deepfake image detection," *Appl. Sci.*, vol. 12, no. 19, p. 9820, Sep. 2022.

[18] J. A. Nasir, O. S. Khan, and I. Varlamis, "Fake news detection: A hybrid CNN-RNN based deep learning approach," *Int. J. Inf. Manag. Data Insights*, vol. 1, no. 1, Apr. 2021, Art. no. 100007.

[19] S. Sheikhi, "An effective fake news detection method using WOA-XgbTree algorithm and content-based features," *Appl. Soft Comput.*, vol. 109, Sep. 2021, Art. no. 107559.

[20] L. Ying, H. Yu, J. Wang, Y. Ji, and S. Qian, "Multi-level multi-modal cross-attention network for fake news detection," *IEEE Access*, vol. 9, pp. 132363–132373, 2021.

[21] Y. Dou, K. Shu, C. Xia, P. S. Yu, and L. Sun, ''User preference-aware fake news detection,'' in *Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2021, pp. 2051–2055.

[22] A. Mosallanezhad, M. Karami, K. Shu, M. V. Mancenido, and H. Liu, ''Domain adaptive fake news detection via reinforcement learning,'' in *Proc. ACM Web Conf.*, Apr. 2022, pp. 3632–3640.

[23] Y. Zhu, Q. Sheng, J. Cao, S. Li, D. Wang, and F. Zhuang, ''Generalizing to the future: Mitigating entity bias in fake news detection,'' in *Proc. 45th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2022, pp. 2120–2125.

[24] S. Imaduwage, P. P. N. V. Kumara, and W. J. Samaraweera, ''Importance of user representation in propagation network-based fake news detection: A critical review and potential improvements,'' in *Proc. 2nd Int. Conf. Adv. Res. Comput. (ICARC)*, Feb. 2022, pp. 90–95.

[25] C. Song, K. Shu, and B. Wu, ''Temporally evolving graph neural network for fake news detection,'' *Inf. Process. Manag.*, vol. 58, no. 6, Nov. 2021, Art. no. 102712.

[26] P. K. Verma and P. Agrawal, ''PropFND: Propagation based fake news detection,'' in *Applications of Artificial Intelligence and Machine Learning*. Berlin, Germany: Springer, 2022, pp. 557–568.

[27] L. Tan, G. Wang, F. Jia, and X. Lian, ''Research status of deep learning methods for rumor detection,'' *Multimedia Tools Appl.*, vol. 82, no. 2, pp. 2941–2982, Jan. 2023.

[28] Y. Wang, L. Wang, Y. Yang, and T. Lian, ''SemSeq4FD: Integrating global semantic relationship and local sequential order to enhance text representation for fake news detection,'' *Exp. Syst. Appl.*, vol. 166, Mar. 2021, Art. no. 114090.

[29] W. Yu, J. Ge, Z. Yang, Y. Dong, Y. Zheng, and H. Dai, ''Multi-domain fake news detection for history news environment perception,'' in *Proc. IEEE 17th Conf. Ind. Electron. Appl. (ICIEA)*, Dec. 2022, pp. 428–433.

[30] C.-V. N. Thi, T.-T. Vuong, D.-T. Le, and Q.-T. Ha, ''v3MFND: A deep multi-domain multimodal fake news detection model for Vietnamese,'' in *Intelligent Information and Database Systems*, Ho Chi Minh City, Vietnam: Springer, 2022, pp. 608–620.

[31] Y. Zhu, Q. Sheng, J. Cao, Q. Nan, K. Shu, M. Wu, J. Wang, and F. Zhuang, ''Memory-guided multi-view multi-domain fake news detection,'' *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 7, pp. 7178–7191, Jul. 2023.

[32] L. Fu, H. Peng, and S. Liu, ''KG-MFEND: An efficient knowledge graph-based model for multi-domain fake news detection,'' *J. Supercomput.*, vol. 37, pp. 1–28, 1987.

[33] M. Sun, X. Zhang, J. Zheng, and G. Ma, ''DDGCN: Dual dynamic graph convolutional networks for rumor detection on social media,'' in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 4, pp. 4611–4619.

[34] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, ''BERT: Pre-training of deep bidirectional transformers for language understanding,'' 2018, *arXiv:1810.04805*.

[35] M. M. Sadiq Fareed, A. Raza, N. Zhao, A. Tariq, F. Younas, G. Ahmed, S. Ullah, S. F. Jillani, I. Abbas, and M. Aslam, ''Predicting divorce prospect using ensemble learning: Support vector machine, linear model, and neural network,'' *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–15, Jul. 2022.

[36] J. Li, J. Wang, M. Xu, and C. Wang, ''Twitter sentiment analysis based on leap LSTM-CNN network,'' *Comput. Simul.*, vol. 38, no. 8, pp. 478–481&486, 2021.

[37] Y. Zhu, Y. Liu, R. Xie, F. Zhuang, X. Hao, K. Ge, X. Zhang, L. Lin, and J. Cao, ''Learning to expand audience via meta hybrid experts and critics for recommendation and advertising,'' in *Proc. 27th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2021, pp. 4005–4013.

[38] Y. Kim, ''Convolutional neural networks for sentence classification,'' in *Proc. Conf. Empir. Method Nat. Lang. Process. (EMNLP)*, Doha, Qatar, 2014, pp. 1746–1751. [Online]. Available: https://aclanthology.org/D14-1181

[39] N. Kalchbrenner, E. Grefenstette, and P. Blunsom, ''A convolutional neural network for modelling sentences,'' 2014, *arXiv:1404.2188*.

[40] R. Johnson and T. Zhang, ''Deep pyramid convolutional neural networks for text categorization,'' in *Proc. 55th Annu. Meeting Assoc. Comput. Linguistics*, vol. 1, 2017, pp. 562–570.

[41] Y. Wang, F. Ma, Z. Jin, Y. Yuan, G. Xun, K. Jha, L. Su, and J. Gao, ''EANN: Event adversarial neural networks for multi-modal fake news detection,'' in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 849–857.

[42] J. Ma, Z. Zhao, X. Yi, J. Chen, L. Hong, and E. H. Chi, ''Modeling task relationships in multi-task learning with multi-gate mixture-of-experts,'' in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2018, pp. 1930–1939.

[43] Z. Qin, Y. Cheng, Z. Zhao, Z. Chen, D. Metzler, and J. Qin, ''Multitask mixture of sequential experts for user activity streams,'' in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 3083–3091.

[44] A. Silva, L. Luo, S. Karunasekera, and C. Leckie, ''Embracing domain differences in fake news: Cross-domain fake news detection using multimodal data,'' in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 1, pp. 557–565.

[45] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, ''Fake news detection on social media: A data mining perspective,'' *ACM SIGKDD Explor. Newslett.*, vol. 19, no. 1, pp. 22–36, 2017.

[46] Q. Nan, D. Wang, Y. Zhu, Q. Sheng, Y. Shi, J. Cao, and J. Li, ''Improving fake news detection of influential domain via domain- and instance-level transfer,'' 2022, *arXiv:2209.08902*.

DAOKANG WANG received the B.S. degree from Shangqiu Normal University, Shangqiu, China, in 2019. He is currently pursuing the master's degree with the School of Electrical and Information Engineering, Hubei University of Automotive Technology. His research interests include deep learning and natural language processing.

WUBO ZHANG received the M.S. degree in computer application technology from the Wuhan University of Science and Technology, Wuhan, China, in 2004. He was a Visiting Scholar with the Beijing Institute of Technology, in 2008. Currently, he is an Associate Professor with the Hubei University of Automotive Technology, Shiyan, China. His research interests include natural language processing, intelligent data analysis, and automobile industry informatization.

WENHUAN WU received the Ph.D. degree from the Xi'an University of Technology, Xi'an, China, in 2020. Currently, he is an Associate Professor with the Hubei University of Automotive Technology, Shiyan, China. His research interests include computer vision, pattern recognition, and image processing.

XIAOLEI GUO received the B.S. degree from Henan University, Kaifeng, China, in 2019. He is currently pursuing the master's degree with the School of Electrical and Information Engineering, Hubei University of Automotive Technology. His research interests include deep learning and natural language processing.